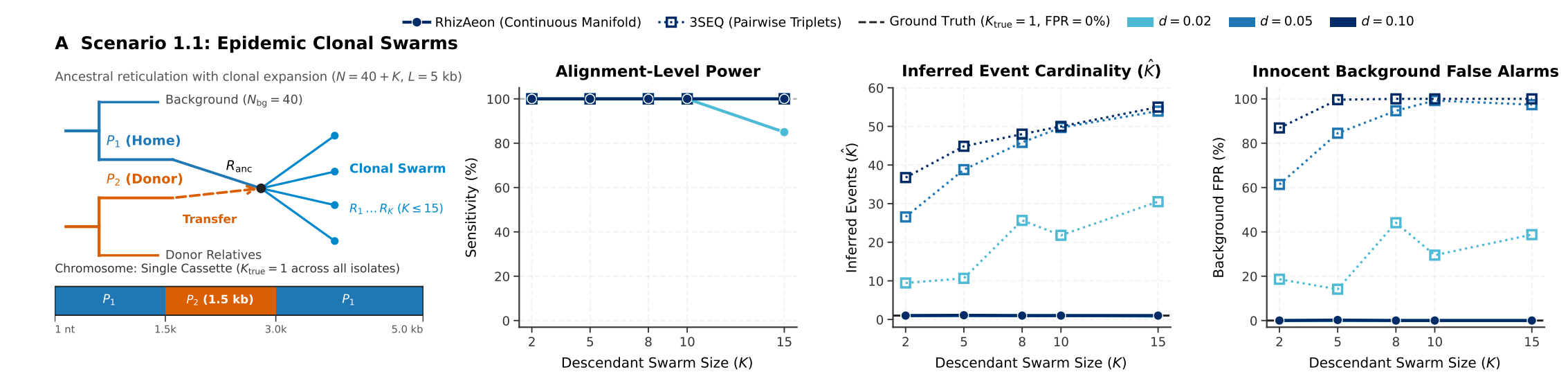
RhizAeon (Continuous Manifold)
3SEQ (Pairwise Triplets)
Ground Truth ($K_{true} = 1$, FPR = 0%)
$d = 0.02$
$d = 0.05$
$d = 0.10$
A Scenario 1.1: Epidemic Clonal Swarms
Ancestral reticulation with clonal expansion ($N = 40 + K$, $L = 5$ kb)
Background ($N_{bg} = 40$)
P_1 (Home)
P_2 (Donor)
R_{anc}
Clonal Swarm
$R_1 \dots R_K$ ($K \leq 15$)
Transfer
Donor Relatives
Chromosome: Single Cassette ($K_{true} = 1$ across all isolates)
P_1
P_2 (1.5 kb)
P_1
1 nt
1.5k
3.0k
5.0 kb
Alignment-Level Power
Sensitivity (%)
Descendant Swarm Size (K)
Inferred Event Cardinality ($\hat{K}$)
Inferred Events ($\hat{K}$)
Descendant Swarm Size (K)
Innocent Background False Alarms
Background FPR (%)
Descendant Swarm Size (K)